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MAKERERE UNIVERSITY BUSINESS SCHOOL

FINAL EXAMINATION FOR THE DEGREE OF BACHELOR OF BUSINESS COMPUTING OF MAKERERE UNIVERSITY ACADEMIC YEAR 2019/2020

COURSE NAME: SYSTEMS ANALYSIS AND DESIGN
COURSE CODE: BUC 2101
SEMESTER: 1.

YEAR OF STUDY: 2
DATE: 28th November 2
TIME: 2:00 PM – 5:00 P

Answer ALL questions from section A and any THREE questions from section B

SECTION A (ANSWER ALL THE QUESTIONS)

Question 1

a. Explain any five qualities of a modern systems analyst (5 marks)
b. Explain any three factors that must be taken into consideration when developing information system (3 Marks)

Question 2

Explain the following phases of information systems development.

a. Problem Analysis Phase *(4 marks)*
b. Design Phase *(4 marks)*

Question 3

a. What roles do system designers and system builders play in information systems development project? (5 Marks)
b. Explain the term "Problem" in three different perspectives in information systems development (3 marks)

Question 4

Question 4

a. Briefly describe software project lifecycle (4 marks)

b. What are the roles of a software development project manager (4 marks)

Question 5

Explain any four responsibilities of an ICT Steering Committee in Software development pr

(8 Marks)

SECTION B (ANSWER ANY THREE QUESTIONS)

Question 6

a) The PIECES framework has been used in determining requirements during system development. Discuss this framework and how it is applied in system development.

(10 marks)

b) In data modelling there is a relationship between the physical data model, logical dat model and conceptual data model, with each one of them having specific features. I following table indicate with a tick or cross which specific feature(s) applies to each the data model. (*a wrong guess or tick will penalise you with 0.5 marks, if you are n*

b) In data modelling there is a relationship between the physical data model, logical dat model and conceptual data model, with each one of them having specific features. I following table indicate with a tick or cross which specific feature(s) applies to each the data model. *(a wrong guess or tick will penalise you with 0.5 marks, if you are n sure of the correct answer don't tick)* **(10 marks)**

FEATURE	PHYSICAL DATA MODEL	LOGICAL DATA MODEL	CONCEPTUAL DATA MODEL
Attributes		X	X
Column data types	X		
Business names	X		
Primary keys	X	X	
Entity names		X	X
Column names	X		
Compound keys	X	X	
Foreign keys	X	X	
Table names	X		
Entity relationships		X	X

Question 7

a) Using an illustration, explain the role played by the primary key and foreign key in th development of a relational database. (5 marks) (illustration 2 marks, explanation marks)

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b) Explain the different tools or techniques that can be used to establish the requirement an information system. (5 marks)

c) In designing a database, there are various processes that must be adopted to have a gc database design. Discuss these design processes of a database. (10 mark:

Question 8

a) With examples, describe the different requirements that must be established by the S Analyst before an information system is developed. (5 marks)

b) With illustrations, explain the following system development methods (5 marl

a) Prototyping
b) V-model
c) Phased development

Question 9

a) What is meant by feasibility study in systems development?
b) Discuss with examples the different types of feasibility studies that a syste normally emphasizes in systems development. (15 Marks)

Question 10

a. Describe the structure of kozar's requirements model in terms of mission, goal objectives, tactics, and information system (10 marks)
b. Use a real world example to explain how kozar's software requirements model applied to elicit and develop software requirements for a business problem (10

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